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Inventor(s)	Lyons; Diane

Method and apparatus for supporting surgical instruments

Abstract

A surgical instrument stand apparatus for supporting and organizing surgical instruments. Apparatus includes a first foot, a second foot, a support tube that extends between the first foot and the second foot and holds the first and second feet at a fixed distance from one another by the support tube, and at least one stand accessory that is selectively operably engageable with the first foot and the second foot for supporting at least one set of surgical instruments. When the at least one set of surgical instruments is supported by the support tube and the at least one stand accessory, the at least one stand accessory is engaged with the first and second feet at first positions. Apparatus may also include at least another stand accessory that is selectively operably engageable with the first and second feet at second positions to support another set of surgical instruments.

Inventors: **Lyons; Diane (North Lawrence, OH)**

Applicant: **Dieble Surgical LLC** (North Lawrence, OH)

Family ID: **95657883**

Assignee: **Dieble Surgical LLC (North Lawrence, OH)**

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Primary Examiner: Novosad; Jennifer E.

Attorney, Agent or Firm: Sand, Sebolt & Wernow Co., LPA

Background/Summary

TECHNICAL FIELD

(1) This disclosure is directed to a surgical instrument stand apparatus for supporting at least one or more types of surgical instruments for surgical or medical procedures.

BACKGROUND ART

(2) Surgical instruments that may be needed during a surgical procedure are conventionally assembled in advance of the surgery on a tray by, for example, a surgery technician or nurse. The tray is stably supported, such as along one side thereof, on rollers above the surgery floor in order to allow mobility and accessibility to the surgical instruments. The height of the tray above the surgery floor is such as to permit the tray to be selectively positioned at will over any portion of an occupied surgery bed, within easy reach of medical personnel. The tray, in combination with the support and rollers therefor, is conventionally referred to as a “mayo stand.”

(3) Various types of surgical instruments are routinely loaded onto a mayo stand in preparation for use in surgery. These surgical instruments are generally arranged on the mayo stand in a nesting, parallel relationship with the handles of the surgical instruments in alignment. In many cases, similar types of surgical instruments are grouped together and graded by size. Such a lineup of surgical instruments is commonly known or referred to as a “stringer” or “stringer of surgical instruments.” Examples of surgical instruments that would commonly be included in a stringer on a mayo stand include hemostats, tonsils, Haney clamps, and needle holders.

(4) However, these types of arrangements on medical stands or mayo stands pose issues to medical personal during surgical or medical procedures. In one instance, surgical instruments that are provided on rolled towels during surgical procedures tend to lean over on these rolled towels which create a greater footprint and/or space on medical stands making it difficult to retrieve or return instruments due to the close proximity of these surgical instruments. In this same instance, the greater footprint and/or space created by the surgical instruments may make retrieving and returning surgical instruments be more difficult or problematic when a specific surgical procedure becomes fast paced. While such retrieval and return issues may be minimal at times, continuous issues of retrieving and returning various types of surgical instruments during a surgical procedure may increase the time of surgery, unnecessary stress and anxiety placed on medical staff during surgery, and potential errors or mistakes in retrieving a desired surgical instrument.

SUMMARY OF THE INVENTION

(5) The presently disclosed surgical instrument stand apparatus accomplishes issues in supporting at least one or more types of surgical instruments for surgical or medical procedures. In one aspect, the surgical instrument stand apparatus includes at least one or a first stand accessory for supporting at least one set of surgical instruments when transporting the surgical instrument stand and the at least one set of surgical instruments. In another aspect, the surgical instrument stand apparatus includes at least another or a second stand accessory for supporting and organizing at least another set of surgical instruments during a surgical or medical procedure. In yet another aspect, the surgical instrument stand apparatus includes at least one expansion assembly that expands and/or extends the overall footprint of the surgical instrument stand apparatus for supporting additional surgical instruments. The presently disclosed surgical instrument stand apparatus addresses inadequacies of known surgical instrument stand apparatuses and devices.

(6) In one aspect, an exemplary embodiment of the present disclosure may provide a surgical instrument stand apparatus. The apparatus includes a first foot, a second foot, and a support tube that extends between the first foot and the second foot, wherein the first foot and the second foot are held at a fixed distance from one another by the support tube. The apparatus also includes at least one stand accessory that is selectively operably engageable with the first foot and the second foot for supporting at least one set of surgical instruments.

(7) This exemplary embodiment or another exemplary embodiment may further include that when the at least one set of surgical instruments is supported by the support tube and the at least one

stand accessory, the at least one stand accessory is engaged with the first foot and the second foot at first positions. This exemplary embodiment or another exemplary embodiment may further include that each of the first foot and the second foot comprises: an outer side; an inner side facing in an opposite direction relative to the outer side; a central opening extending between the outer side and the inner side and configured to receive the support tube; and at least two accessory openings extending between the outer side and the inner side and being offset from the central opening; wherein one or both of the at least two accessory openings are configured to receive the at least one stand accessory. This exemplary embodiment or another exemplary embodiment may further include at least another stand accessory selectively operably engageable with the first foot and the second foot at second positions different than the first positions, wherein at least another set of surgical instruments is supported by the at least another stand accessory. This exemplary embodiment or another exemplary embodiment may further include that each of the first foot and the second foot further comprises: a bottom end that is configured to engage with a support surface; and a top end that is vertically opposite to the bottom end and is free from engaging the support surface; wherein the at least another stand accessory is configured to selectively operably engage with the top end of each of the first foot and the second foot. This exemplary embodiment or another exemplary embodiment may further include that the first foot further comprises: a first slit extending downwardly into the first foot from the top end towards the bottom end, wherein the first slit is configured to receive a first end of the at least another stand accessory; and a second slit extending downwardly into the second foot from the top end towards the bottom end and being coaxial with the first slit of the first foot, wherein the second slit is configured to receive a second end of the at least another stand accessory that is opposite to the first end of the at least another stand accessory. This exemplary embodiment or another exemplary embodiment may further include that the support tube comprises: a first end that operably engages with the first foot; a second end that is longitudinally opposite to the first end and operably engages with the second foot; and a wall extending between the first end and the second end and defining a passageway therethrough with an inner diameter. This exemplary embodiment or another exemplary embodiment may further include that the support tube further comprises: at least one opening defined in the wall between the first end of the support tube and the second end of the support tube; wherein the at least one opening provides fluid communication between the passageway of the support tube and an external environment that surrounds the support tube. This exemplary embodiment or another exemplary embodiment may further include an expansion assembly selectively operably engagable with the support tube at one of the first end of the support tube and the second end of the support tube. This exemplary embodiment or another exemplary embodiment may further include that the expansion assembly comprises: an expansion foot; and an expansion support bar extending between the expansion foot and one of the first foot and the second foot. This exemplary embodiment or another exemplary embodiment may further include that the expansion support bar comprises: a first end that operably engages with the expansion foot; a second end that is longitudinally opposite to the first end and is configured to selectively operably engage with the support tube at one of the first end of the support tube and the second end of the support tube; and a wall extending between the first end and the second end and defining an outer diameter that is less than the inner diameter of the support tube; wherein the expansion support bar is slidably moveably inside of the passageway of the support tube. This exemplary embodiment or another exemplary embodiment may further include that the expansion foot comprises: a bottom expansion end that is configured to engage with a support surface; a top end that is vertically opposite to the bottom end and is free from engaging the support surface; and an expansion slit extending downwardly into the expansion foot from the top expansion end towards the bottom expansion end and is coaxial with a first slit defined in the first foot and a second slit defined in the second foot; wherein at least another stand accessory is configured to selectively operably engage with the top end of the expansion foot inside of the expansion slit remote from the expansion support bar. This exemplary embodiment or another exemplary embodiment may further include that the at least another stand accessory comprises: at least one set of recesses defined in the at least another stand accessory; wherein the at least another set of surgical instruments is held by the at least another stand accessory inside the at least one set of recesses. This exemplary embodiment or another exemplary embodiment may further include at least one clip that is selectively operably engageable with the support tube; wherein the at least one clip is selectively positionable along the support tube between the first foot and the second foot, wherein the at least one clip is configured to separate one surgical instrument of the at least one set of surgical instruments from another surgical instrument of the at least one set of surgical instruments, and wherein the at least one clip is operative to hold a surgical instrument of the at least one set of surgical instruments in operative engagement with the support tube in a desired orientation. This exemplary embodiment or another exemplary embodiment may further include at least another stand accessory selectively operably engageable with the support tube between the first foot and the second foot, wherein at least another set of surgical instruments is supported by the at least another stand accessory inside at least one set of recesses defined in the at least another stand accessory. This exemplary embodiment or another exemplary embodiment may further include at least another stand accessory selectively operably engageable with the support tube between the first foot and the second foot, wherein at least another set of surgical instruments is supported by the at least another stand accessory remote from the support tube. This exemplary embodiment or another exemplary embodiment may further include that the at least another stand accessory further comprises: a plate; and a pair of clips that pivotably engage with the plate; wherein the plate and the pair of clips cooperatively engage with the support tube between the first foot and the second foot. This exemplary embodiment or another exemplary embodiment may further include the at least another stand accessory further comprises: at least one set of recesses defined in the plate; wherein the at least another set of surgical instruments is supported by the plate inside of the at least one set of recesses.

(8) In another aspect, an exemplary embodiment of the present disclosure may provide a method of supporting at least one set of surgical instruments with a surgical instrument stand apparatus. The method comprises steps of engaging a first foot of the surgical instrument stand apparatus with a first end of a support tube of the surgical instrument stand apparatus; engaging a second foot of the surgical instrument stand apparatus with a second end of the support tube, wherein the second foot is spaced apart from the first foot at a fixed distance defined by the support tube; selecting between a first stand accessory and a second stand accessory; engaging the selected first stand accessory or the second stand accessory with the first foot and the second foot; and supporting at least one set of surgical instruments with the surgical instrument stand apparatus by the support tube and the selected first stand accessory or the selected second stand accessory.

(9) This exemplary embodiment or another exemplary embodiment may further include that when the first stand accessory is selected, the method further comprises: inserting a first end of the first stand accessory through the first foot, the second foot, and first handles of the at least one set of surgical instruments; engaging the first stand accessory with the first foot, the second foot, and the first handles of the at least one set of surgical instruments; inserting a second end of the first stand accessory through second handles of the at least one set of surgical instruments; and engaging the first stand accessory with the second handles of the at least one set of surgical instruments. This exemplary embodiment or another exemplary embodiment may further include steps of clamping at least one clip with the support tube; and separating the at least one set of surgical instruments into at least two groups. This exemplary embodiment or another exemplary embodiment may further include that when the second stand accessory is selected in the step of selecting between the first stand accessory and the second stand accessory, the method further comprises: engaging a first end of the second stand accessory with the first foot inside a first slit defined in the first foot; and engaging a second end of the second stand accessory with the second foot inside a second slit

defined in the second foot; wherein the second stand accessory is positioned above and spaced apart from the support tube. This exemplary embodiment or another exemplary embodiment may further include a step of engaging at least one expansion assembly with the support tube at one or both of the first end of the support tube and the second end of the support tube. This exemplary embodiment or another exemplary embodiment may further include that when the second stand accessory is selected in the step of selecting between the first stand accessory and the second stand accessory, the method further comprises: engaging a first end of the second stand accessory with one of the first foot inside a first slit defined in the first foot and the second foot inside a second slit defined in the second foot; and engaging a second end of the second stand accessory with an expansion foot of the expansion assembly inside an expansion slit defined in the expansion foot; wherein the second stand accessory is positioned above and spaced apart from the support tube and an expansion support bar of the expansion assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Sample embodiments of the present disclosure are set forth in the following description, are shown in the drawings and are particularly and distinctly pointed out and set forth in the appended claims.

(2) FIG. **1** (FIG. **1**) is a front, top, first side isometric perspective view of a surgical instrument stand apparatus in accordance with one aspect of the present disclosure, wherein a set of first surgical instruments is supported by the surgical instrument stand apparatus with a first stand accessory and a set of clips.

(3) FIG. **2** (FIG. **2**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus shown in FIG. **1**.

(4) FIG. **3** (FIG. **3**) is an exploded view of the surgical instrument stand apparatus shown in FIG. **1**.

(5) FIG. **4** (FIG. **4**) is a first side elevation view of a first foot of the surgical instrument stand apparatus shown in FIG. **1**.

(6) FIG. **5** (FIG. **6**) is a front elevation view of the first foot of the surgical instrument stand apparatus shown in FIG. **1**.

(7) FIG. **6** (FIG. **6**) is an operational view of the surgical instrument stand apparatus being placed on a surgical table while holding the set of first surgical instruments and the first stand accessory being removed from the surgical instrument stand apparatus

(8) FIG. **7** (FIG. **7**) is another operational view of the surgical instrument stand apparatus being sterilized in a sterilization apparatus while holding the set of first surgical instruments with the first stand accessory and the set of clips.

(9) FIG. **8** (FIG. **8**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus in accordance with another aspect of the present disclosure, wherein a first group of a set of first surgical instruments and a second group of a set of first surgical instruments are supported by the surgical instrument stand apparatus with first stand accessories and a set of clips.

(10) FIG. **9** (FIG. **9**) is a rear, top, first side isometric perspective view of the surgical instrument stand apparatus shown in FIG. **8**, wherein the first stand accessories are removable from the set of first surgical instruments.

(11) FIG. **10** (FIG. **10**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus in accordance with another aspect of the present disclosure, wherein at least one expansion assembly is engaged with a support tube of the surgical instrument stand apparatus.

(12) FIG. **11** (FIG. **11**) is a cross-sectional view of the support tube of the surgical instrument stand apparatus and an expansion support of the expansion assembly taken in the direction of line **11-11**

as shown in FIG. **10**; wherein the expansion support engages with the support tube inside of the support tube.
(13) FIG. **12** (FIG. **12**) is an exploded view of the expansion assembly shown in FIG. **10**.
(14) FIG. **13** (FIG. **13**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus in accordance with another aspect of the present disclosure, wherein a set of second surgical instruments is supported by a second stand accessory that is equipped to the surgical instrument stand apparatus.
(15) FIG. **14** (FIG. **14**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus equipped with the second stand accessory as shown in FIG. **13**.
(16) FIG. **15** (FIG. **15**) is an exploded view of the surgical instrument stand apparatus and the second stand accessory.
(17) FIG. **16** (FIG. **16**) is a front elevation view of the second stand accessory.
(18) FIG. **17** (FIG. **17**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus in accordance with another aspect of the present disclosure, wherein the set of first surgical instruments is supported with the expansion assembly and at least one clip, and wherein a set of second surgical instruments is supported by a second stand accessory equipped to the surgical instrument stand apparatus.
(19) FIG. **18** (FIG. **18**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus shown in FIG. **17**, wherein the surgical instrument stand apparatus is equipped with at least one expansion assembly and the second stand accessory.
(20) FIG. **19** (FIG. **19**) is a front elevation view of an alternative second stand accessory.
(21) FIG. **20** (FIG. **20**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus in accordance with another aspect of the present disclosure, wherein the surgical instrument stand apparatus is equipped with a first expansion assembly, a second expansion assembly, and the second stand accessory.
(22) FIG. **21** (FIG. **21**) is a front, top, first side isometric perspective view of the surgical instrument stand apparatus in accordance with another aspect of the present disclosure, wherein the surgical instrument stand apparatus is equipped with a clamping rack.
(23) FIG. **22** (FIG. **22**) is a front, top, first side isometric perspective view of the clamping rack shown in FIG. **21**.
(24) FIG. **23** (FIG. **23**) is a rear, top, second side isometric perspective view of a clip of the clamping rack shown in FIG. **21**.
(25) FIG. **24** (FIG. **24**) is an exploded view of the clamping rack shown in FIG. **21**.
(26) FIG. **25** (FIG. **25**) is a sectional view taken in the direction of line **25-25** shown in FIG. **21**.
(27) FIG. **26** (FIG. **26**) is an exemplary kit of the surgical instrument stand apparatus, a set of clips, a pair of first stand accessories, and at least one second stand accessory.
(28) FIG. **27** (FIG. **27**) is an exemplary method flowchart.
(29) Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

(30) FIGS. **1-7** illustrate a first embodiment or configuration of a surgical instrument stand apparatus (hereinafter "stand") generally referred to as **1**. As illustrated in FIGS. **1-7**, stand **1** is configured to support and organize at least one set of surgical instruments that is selected by medical professions for a particular surgical procedure or medical procedure. It should be understood that stand **1** assists medical professions in organizing and arranging various types of surgical instruments along the stand **1** for easily transporting these surgical instruments from a preparation station to a surgical or operating room or to select a desired surgical instrument during a surgical procedure. The components and devices that form the stand **1** are now discussed in greater detail below.
(31) Stand **1** includes a first foot or upright support **10**. As best seen in FIGS. **4-5**, the first foot **10** includes a first or front end **10**A, a second or rear end **10**B longitudinally opposite to the front end

10A, and a longitudinal direction defined therebetween. As best seen in FIG. **5**, first foot **10** also includes a first or outer side **10**C that extends between the front end **10**A and the rear end **10**B, a second or inner side **10**D that extends between the front end **10**A and the rear end **10**B and is transversely opposite to the outer side **10**C, and a transverse direction defined therebetween. As best seen in FIGS. **4-5**, first foot **10** also includes a top end **10**E that is positioned above the front end **10**A, the rear end **10**B, the outer side **10**C, and the inner side **10**D, a bottom end **10**F that is positioned below the front end **10**A, the rear end **10**B, the outer side **10**C, and the inner side **10**D and is vertically opposite to the top end **10**E, and a vertical direction defined between. First foot **10** also defines an overall height **10**G that is measured between the top end **10**E and the bottom end **10**F (see FIG. **5**). First foot **10** also defines an overall width **10**H that is measured between the outer side **10**C and the inner side **10**D and is less than the overall height **10**G in the present disclosure (see FIG. **5**). In other exemplary embodiments, first foot **10** may define any suitable heights, widths, or lengths dictated by the implementation of the stand **1** in surgical and/or medical procedures.

(32) Still referring to first foot **10**, first foot **10** also defines a central opening **10**J. As best seen in FIG. **4**, the central opening **10**J extends entirely through the first foot **10** along the transverse direction of the first foot **10** between the outer side **10**C and the inner side **10**D. In the present disclosure, the outer side **10**C and the inner side **10**D are in operative communication with one another at the central opening **10**J. In the present disclosure, the central opening **10**J is also an oblong-shaped or oval-shaped opening that is configured to receive an end of a support tube of the stand **1** to engage the first foot **10** and the support tube with one another, which is discussed in greater detail below. In other exemplary embodiment, central opening **10**J may define any suitable shape that is configured to receive an end of a support tube of the stand **1** to engage the first foot **10** and the support tube with one another.

(33) Still referring to first foot **10**, first foot **10** also defines a pair of accessory openings **10**K. As best seen in FIG. **4**, each accessory opening of the pair of accessory openings **10**K extends entirely through the first foot **10** along the transverse direction of the first foot **10** between the outer side **10**C and the inner side **10**D. In the present disclosure, the outer side **10**C and the inner side **10**D are in operative communication with one another at each accessory opening of the pair of accessory openings **10**K. The pair of accessory openings **10**K is also spaced apart and offset from the central opening **10**J and defined proximate to the rear end **10**B of first foot **10**. In the present disclosure, a first accessory opening **10**K**1** of the pair of the accessory openings **10**K is defined below a second accessory opening **10**K**2** of the pair of the accessory openings **10**K.

(34) Each accessory opening of the pair of accessory openings **10**K also defines a pair of axes **10**L that is parallel with the transverse direction of the first foot **10**. Upon assembly, an instrument stringer or first stand accessory of the stand **1** may lie on one of axes of the pair of axes **10**L for supporting and organizing surgical instruments along the length of the stand **1**, which are discussed in greater detail below. Each axis of the pair of axes **10**L is also offset from one another and defined at an angle **10**M measured between a first axis **10**L**1** of the pair of axes **10**L and a second axis **10**L**2** of the pair of axes **10**L. In the present disclosure, the angle **10**M measured between the pair of axes **10**L is an acute angle. With such configuration, each accessory opening of the pair of accessory opening **10**K is also angled and/or slanted based on the angle **10**M measured between the pair of axes **10**L extending through each accessory opening of the pair of accessory opening **10**K.

(35) In the present disclosure, each accessory opening of the pair of accessory openings **10**K is an oblong-shaped or oval-shaped opening that is configured to receive an instrument stringer or first stand accessory of the stand **1** for supporting and organizing surgical instruments along the length of the stand **1**, which are discussed in greater detail below. In other exemplary embodiment, each accessory opening of the pair of accessory openings **10**K may define any suitable shape that is configured to receive an instrument stringer or first stand accessory of the stand **1** for supporting and organizing surgical instruments along the length of the stand **1**.

(36) While first foot **10** defines the pair of accessory openings **10**K, a first foot or any foot discussed and illustrated herein may define any suitable number of accessory openings as dictated by the implementation of the accessory openings, including the number of instrument stringers or first stand accessories that are intended to be received by a first foot or any foot of a stand mentioned herein. Examples of suitable numbers of accessory openings defined in a first foot or any foot discussed and illustrated herein include at least one, two, a plurality, three, four, five, and other suitable numbers of accessory openings that may be defined in a foot discussed and illustrated herein. In one exemplary embodiment, a first set of accessory openings may be defined at a front end of a foot, and a second set of accessory openings may be defined at a rear end of the foot opposite to the first set of accessory openings.
(37) Still referring to first foot **10**, first foot **10** also defines a first slit **10**N. As best in FIG. **4**, the first slit **10**N is defined by a pair of interior vertical walls **10**N**1** that extends vertically downward into the first foot **10** from the top end **10**E to an interior base wall **10**N**2**. The first slit **10**N also defines a slit height **10**P that is measured along the lengths of the pair of interior vertical walls **10**N**1** from the top end **10**E to the interior base wall **10**N**2**. The first slit **10**N also defines a slit width **10**Q that is measured between the pair of interior vertical walls **10**N**1**. Such use of the first slit **10**N is discussed in greater detail below.
(38) Stand **1** also includes a second foot **20**. In the present disclosure, the first foot **10** and the second foot **20** are identical to one another for stand **1**. As such, a front end **20**A, a rear end **20**B, an outer side **20**C, an inner side **20**D, a top end **20**E, a bottom end **20**F, a central opening **20**J, a pair of accessory openings **20**K, and a second slit **20**N defined by a pair of internal vertical walls **20**N**1** and an internal base wall **20**N**2** of the second foot **20** are identical to the front end **10**A, rear end **10**B, outer side **10**C, inner side **10**D, top end **10**E, bottom end **10**F, central opening **10**J, pair of accessory openings **10**K, and the first slit **10**N defined by the pair of internal vertical walls **10**N**1** and the internal base wall **10**N**2** of the first foot **10**. It should be understood that, while not illustrated herein, the overall height **10**G and the overall width **10**H of the first foot **10** applies equally to an overall height and overall width of the second foot **20**. It should also be understood that, while not illustrated herein, the pair of axes **10**L and the angle **10**M of the pair of accessory openings **10**K of the first foot **10** applies equally to the pair of accessory openings **20**K of the second foot **20**. It should also be understood that, while not illustrated herein, the interior vertical walls **10**N**1** and the internal base wall **10**N**2** that define the first slit **10**N having the slit height **10**P and the slit width **10**Q applies equally to the second slit **20**N of the second foot **20**.
(39) While second foot **20** defines the pair of accessory openings **20**K, a second foot or any foot discussed and illustrated herein may define any suitable number of accessory openings as dictated by the implementation of the accessory openings, including the number of instrument stringers or first stand accessories that are intended to be received by a second foot or any foot of a stand mentioned herein. Examples of suitable numbers of accessory openings defined in a second foot or any foot discussed and illustrated herein include at least one, two, a plurality, three, four, five, and other suitable numbers of accessory openings that may be defined in a foot discussed and illustrated herein. In one exemplary embodiment, a first set of accessory openings may be defined at a front end of a foot, and a second set of accessory openings may be defined at a rear end of the foot opposite to the first set of accessory openings.
(40) Stand **1** also includes a support tube **30** that operably engages with the first foot **10** and the second foot **20**. As best seen in FIG. **3**, the support tube **30** includes a first end **30**A, a second end **30**B that is longitudinally opposite to the first end **30**A, a wall **30**C that extends longitudinally between the first end **30**A and the second end **30**B, and a longitudinal direction defined therebetween. The support tube **30** also includes an exterior surface **30**D that extends along the entire length of the wall **30**C between the first end **30**A and the second end **30**B. The support tube **30** also includes an interior surface **30**E that extends along the entire length of the wall **30**C between the first end **30**A and the second end **30**B and faces in an opposite direction relative to the

exterior surface **30**D.
(41) Still referring to support tube **30**, support tube **30** also defines a passageway **30**F. As best seen in FIGS. **3** and **11**, the passageway **30**F extends between the first end **30**A and the second end **30**B along the interior surface **30**E of the support tube **30**. The passageway **30**F is also accessible at the first end **30**A and the second end **30**B since the first end **30**A and the second end **30**B are open ends. Support tube **30** also includes a set of internal ridges **30**G that extends outwardly from the interior surface **30**E and into the passageway **30**F. In the present disclosure, each interior ridge of the set of internal ridges **30**G also runs along the entire length of the support tube **30** between the first end **30**A and the second end **30**B; such use and purpose of the set of internal ridges **30**G is discussed in greater detail below.
(42) Still referring to support tube **30**, support tube **30** also defines a set of apertures **30**H. As best seen in FIG. **3**, each aperture of the set of apertures **30**H extends through the wall **30**C from the exterior surface **30**D to the interior surface **30**E; the exterior surface **30**D and the interior surface **30**E are in operative communication with one another at each aperture of the set of apertures **30**H. Each aperture of the set of apertures **30**H also provides fluid communication between the passageway **30**F of the support tube **30** and the exterior environment that surrounds the support tube **30**. Such communication between the passageway **30**F of the support tube **30** and the exterior environment may provide various uses for stand **1**, including sterilizing the interior spaces of support tube **30** after being used in a surgical procedure, viewing a position of an expansion bar of an expansion assembly inside of the support tube **30**, and other various uses that will be discussed in greater detail below.
(43) In the present disclosure, the support tube **30** defines two apertures **30**H. While support tube **30** defines two apertures **30**H, a support tube discussed and illustrated herein may define any suitable number of apertures as dictated by the implementation of the accessory openings, including the number of apertures to allow sterilization and/or cleaning solution to enter into interior spaces of support tube **30** after being used in a surgical procedure, positions to view an expansion bar of an expansion assembly inside of the support tube **30**, and other various uses. Examples of suitable numbers of apertures defined in a support tube include at least one, two, a plurality, three, four, five, and other suitable numbers of apertures that may be defined in a support tube discussed and illustrated herein.
(44) Still referring to support tube **30**, support tube **30** may also include a set of projections **30**J. As best seen in FIG. **3**, a first projection **30**J**1** of the set of projections **30**J is defined at the first end **30**A of the support tube **30**, and a second projection **30**J**2** of the set of projections **30**J is defined at the second end **30**B of the support tube **30** opposite to the first projection **30**J**1**. Such inclusion of the set of projections **30**J may provide attachment means between the first foot **10** and support tube **30** when assembled with one another and between the second foot **20** and support tube **30** when assembled with one another. Stated differently, support tube **30** frictionally fits with the first foot **10** and with the second foot **20** due to the set of projections **30**J pressing against an interior wall defined inside each central opening **10**J, **20**K of the first foot **10** and the second foot **20**.
(45) Still referring to support tube **30**, support tube **30** may also define an inner diameter **30**K and an outer diameter **30**L As best seen in FIG. **11**, the inner diameter **30**K of the support tube **30** is measured along the set of internal ridges **30**G and is continuous from the first end **30**A to the second end **30**B. Referring to FIG. **3**, the outer diameter **30**L of the support tube **30** is also continuous along the support tube **30** from the first end **30**A to the second end **30**B. As best seen in FIG. **2**, the outer diameter **30**L is less than an inner diameter of the central opening **10**J of the first foot **10** and an inner diameter of the central opening **20**J of the second foot **20** so that support tube **30** is received by and engages with the first foot **10** and the second foot **20**.
(46) It should be understood that the support tube **30** may define any suitable length necessary for transporting and organizing surgical and/or medical equipment. In one exemplary embodiment, a length of a support tube discussed herein may be approximately six inches long when measured

between a first end of the support tube and a second end of the support tube. In another exemplary embodiment, another length of a support tube discussed herein may be approximately twelve inches long when measured between a first end of the support tube and a second end of the support tube. In yet another exemplary embodiment, another length of a support tube discussed herein may be between six inches up to about twelve inches when measured between a first end of the support tube and a second end of the support tube. In yet another exemplary embodiment, another length of a support tube discussed herein may be at least twelve inches when measured between a first end of the support tube and a second end of the support tube.
(47) Stand **1** may also be equipped with at least one first stand accessory or stringer bar **40**. As best seen in FIG. **1**, the stringer bar **40** includes a first end **40**A, a second end **40**B opposite to the first end **40**A, and a cylindrical wall **40**C that extends between the first end **40**A and the second end **40**B. The stringer bar **40** also includes a first portion **40**D that extends from the first end **40**A to a second portion **40**E where the first portion **40**D and the second portion **40**E are orthogonal to one another. The stringer bar **40** also includes a third portion **40**F that extends from the second end **40**B to the second portion **40**E where the second portion **40**E and the third portion **40**F are orthogonal to one another. In the present disclosure, the stringer bar **40** is provided in a U-shaped configuration in order to support and maintain handles and/or grips of surgical instruments on the stand **1**. In one exemplary embodiment, any conventional or commercially available stringer or similar device may be equipped with stand **1** for supporting surgical instruments on the stand **1**. Such use of the stringer bar **40** is discussed in greater detail below.
(48) Stand **1** may also be equipped with one or more clips **50**. As best seen in FIG. **1**, clips **50** operably engage with the support tube **30** at any position between the first foot **10** and the second foot **20**. In the present disclosure, two clips **50** are shown being clamped with the support tube **30** to separate and/or organize a set of first surgical instruments **60** into desired groups. In other exemplary embodiments, any suitable number of clips **50** may be clamped with the support tube **30** to separate and/or organize a set of first surgical instruments **60** into a desired number of groups. In one exemplary embodiment, clips **50** discussed and illustrated herein may be clips disclosed in U.S. patent application Ser. No. 12/315,839 with the title “SURGICAL STANDS, SURGICAL INSTRUMENT ORGANIZER ASSEMBLIES, AND METHODS OF USE THEREFOR.” In other exemplary embodiments, clips **50** discussed and illustrated herein may also be conventional and/or commercially available surgical clips.
(49) Having now discussed the components of the stand **1** and other accessories and/or equipment that may be equipped with the stand **1**, methods of assembling stand **1** and methods of using stand **1** with surgical instruments are now discussed in greater detail below.
(50) Prior to introducing a set of first surgical instruments **60** to the stand **1**, a medical professional or user may assemble a single stand **1** (or more than one stand **1** if desired) for a surgical or medical procedure. Initially, the user may grab the first foot **10** and the support tube **30** to begin assembling the stand **1**. Upon assembly, the user introduces the first end **30**A of the support tube **30** into the central opening **10**J of the first foot **10** to engage the first foot **10** and the support tube **30** with one another. The first projection **30**J**1** of the set of projections **30**J may also provide attachment means inside of the central opening **10**J of the first foot **10** to frictionally fit the first foot **10** and the support tube **30** with one another. Subsequently, the user introduces the second end **30**B of the support tube **30** into the central opening **20**J of the second foot **20** to engage the second foot **20** and the support tube **30** with one another. The second projection **30**J**2** of the set of projections **30**J may also provide attachment means inside of the central opening **20**J of the second foot **20** to frictionally fit the second foot **20** and the support tube **30** with one another.
(51) Upon assembly of the stand **1**, the user may then introduce the set of first surgical instruments **60** to the stand **1**. Once introduced, each surgical instrument of the set of first surgical instruments **60** illustrated herein may rest on and be supported by the support tube **30**. The user may then introduce one or more clips **50** to the stand **1** to organize and/or arrange the set of first surgical

instruments **60** into one or more desired groups. As best seen in FIG. **1**, two clips **50** are shown being clamped to the support tube **30** at two locations between the first end **30**A and the second end **30**B to divide and/or separate the set of first surgical instruments into three distinct groups. It should be understood that the user may introduce and clamp the clips **50** with the support tube **30** prior to introducing the set of first surgical instrument **60** to the stand **1**.

(52) Once the clips **50** and the set of first surgical instruments **60** are provided with stand **1**, user may then introduce and temporarily engage a stringer bar **40** with the stand **1** and the set of first surgical instruments **60**. As best seen in FIG. **1**, the first end **40**A of the stringer bar **40** may pass through a second accessory opening **20**K**2** of the pair of accessory openings **20**K of the second foot **20**. The first end **40**A of the stringer bar **40** also passes through the second accessory opening **10**K**2** of the pair of accessory openings **10**K of the first foot **10** given that the second accessory opening **10**K**2** of the pair of accessory openings **10**K of the first foot **10** and the second accessory opening **20**K**2** of the pair of accessory openings **20**K of the second foot **20** are coaxial with one another when stand **1** is assembled. The user continues to pass the first end **40**A of the stringer bar **40** through the first foot **10** and the second foot **20** until the first portion **40**D is supported and engaged with the first foot **10** and the second foot **20** inside of the second accessory opening **10**K**2** of the pair of accessory openings **10**K of the first foot **10** and the second accessory opening **20**K**2** of the pair of accessory openings **20**K of the second foot **20**. As the first end **40**A passes through the first foot **10** and the second foot **20**, the first end **40**A of the stringer bar **40** simultaneously passes through first handles and/or grips **61**A of the set of first surgical instruments **60** so that the first portion **40**D hangs and supports the set of first surgical instruments **60**. As the first end **40**A passes through the first foot **10** and the second foot **20**, the second end **40**B of the stringer bar **40** may simultaneously (or subsequently) pass through second handles and/or grips **61**B of the set of first surgical instruments **60** opposite to the first handles and/or grips **61**A of the set of first surgical instruments **60**. Once the second end **40**B passes through second handles and/or grips **61**B of the set of first surgical instruments **60**, the third portion **40**F supports the set of first surgical instruments **60** to maintain the set of first surgical instruments **60** at an open position. Upon such engagement of the stand **1**, the stringer bar **40**, the clips **50**, and the set of first surgical instruments **60**, the user may then collectively transport the stand **1**, the stringer bar **40**, the clips **50**, and the set of first surgical instruments **60** to desired locations.

(53) In one instance, and as best seen in FIG. **6**, the user may collectively transport the stand **1**, the stringer bar **40**, the clips **50**, and the set of first surgical instruments **60** to a support surface or surgical tray/table (generally referred to as “S” in FIG. **6**) for a surgical or medical procedure. Once the stand **1**, the stringer bar **40**, the clips **50**, and the set of first surgical instruments **60** are resting on the support surface, the user may then remove the stringer bar **40** from the stand **1** and the set of first surgical instruments **60** so users or medical professions may simply retrieve and remove one or more surgical instruments of the set of first surgical instruments from the stand **1**. Such removal of the stringer bar **40** is denoted by an arrow labeled “A” in FIG. **6**. Once the stringer bar **40** is removed, the set of first surgical instruments **60** rests on and is supported by the support tube **30** and the clips **50** until users or medical professions retrieve and remove one or more surgical instruments of the set of first surgical instruments from the stand **1**.

(54) In another instance, and as best seen in FIG. **7**, the user may collectively transport used and/or non-sanitary stand **1**, the stringer bar **40**, the clips **50**, and the set of first surgical instruments **60** to a sterilization apparatus (generally referred to as “SA” in FIG. **7**). In this instance, the used and/or non-sanitary stand **1**, the stringer bar **40**, the clips **50**, and the set of first surgical instruments **60** are placed into the sterilization apparatus for sterilizing and/or cleaning purposes due to the devices and/or apparatuses being used in a previous surgical or medical procedure. It should be understood that a user may reengage the stringer bar **40** with the used stand **1** and the used set of first surgical instruments **60** to maintain the used set of first surgical instruments **60** with the stand **1** during sterilization and/or cleaning procedures. While not illustrated herein, the stand **1** may also be

disassembled such that the first foot **10**, the second foot **20**, and the support tube **30** are placed into a sterilization apparatus that is illustrated herein or another commercially-available sterilizing apparatus for sterilizing and/or cleaning purposes after being used in a surgical or medical procedure.

(55) Stand **1** may also be equipped with an expansion assembly **70**. As best seen in FIG. **10**, the expansion assembly **70** is selectively operably engageable with the support tube **30** at the first end **30**A of the support tube **30** or at the second end **30**B of the support tube **30**. It should be understood that the expansion assembly **70** is also slidably moveable with the support tube **30** to vary the overall length of the stand **1** (includes the first foot **10**, the second foot **20**, and the support tube **30**) and the expansion assembly **70**; such linear movement of the expansion assembly **70** is denoted by a double arrow labeled “B” in FIGS. **10** and **18**. Such components that form the expansion assembly **70** are discussed in greater detail below.

(56) Expansion assembly **70** includes an expansion foot **72**. In the present disclosure, certain features of the expansion foot **72** are identical to certain features of the first foot **10**. As such, a front end **72**A, a rear end **72**B, a first side **72**C, a second side **72**D, a top end **72**E, a bottom end **72**F, a pair of accessory openings **72**K, and an expansion slit **72**N of the expansion foot **72** are identical to the front end **10**A, rear end **10**B, outer side **10**C, inner side **10**D, top end **10**E, bottom end **10**F, pair of accessory openings **10**K, and the first slit **10**N of the first foot **10**. It should be understood that, while not illustrated herein, the overall height **10**G and the overall width **10**H of the first foot **10** applies equally to an overall height and overall width of the expansion foot **72**. It should also be understood that, while not illustrated herein, the pair of axes **10**L and the angle **10**M of the pair of accessory openings **10**K of the first foot **10** applies equally to the pair of accessory openings **72**K of the expansion foot **72**. It should also be understood that, while not illustrated herein, the interior vertical walls **10**N**1** and the internal base wall **10**N**2** that define the first slit **10**N having the slit height **10**P and the slit width **10**Q applies equally to the expansion slit **72**N of the expansion foot **72**.

(57) However, the expansion foot **72** includes different features and/or configurations when compared to the first foot **10**. As best seen in FIG. **12**, the expansion foot **72** includes a central opening **72**J that is similar to the central opening **10**J of the first foot **10**. In this embodiment, however, an inner diameter of the central opening **72**J is less than and/or smaller than an inner diameter of the central opening **10**J of the first foot **10** due to an outer diameter of an expansion support bar of the expansion assembly **70**, which is discussed in greater detail below.

(58) While expansion foot **72** defines the pair of accessory openings **72**K, an expansion foot or any foot discussed and illustrated herein may define any suitable number of accessory openings as dictated by the implementation of the accessory openings, including the number of instrument stringers or first stand accessories that are intended to be received by an expansion foot or any foot of a stand mentioned herein. Examples of suitable numbers of accessory openings defined in an expansion foot or any foot discussed and illustrated herein include at least one, two, a plurality, three, four, five, and other suitable numbers of accessory openings that may be defined in a foot discussed and illustrated herein. In one exemplary embodiment, a first set of accessory openings may be defined at a front end of an expansion foot, and a second set of accessory openings may be defined at a rear end of the expansion foot opposite to the first set of accessory openings.

(59) Expansion assembly **70** also includes an expansion support bar **74** that operably engages with the support tube **30** and the expansion foot **72**. As best seen in FIG. **12**, the expansion support bar **74** includes a first end **74**A, a second end **74**B that is longitudinally opposite to the first end **74**A, a wall **74**C that extends longitudinally between the first end **74**A and the second end **74**B, and a longitudinal direction defined therebetween. The expansion support bar **74** also includes an exterior surface **74**D that extends along the entire length of the wall **74**C between the first end **74**A and the second end **74**B. The expansion support bar **74** also defines an outer diameter **74**E that is continuous along the wall **74**C between the first end **74**A and the second end **74**B. In the present

disclosure, the outer diameter **74**E of the expansion support bar **74** is less than the inner diameter **30**K of the support tube **30** so that the expansion support bar **74** is received by and engages with the support tube **30** inside of the passageway **30**F; such assembly of the expansion assembly with the stand **1** is discussed in greater detail below.
(60) Still referring to expansion support bar **74**, expansion support bar **74** may also include a projection **74**F. As best seen in FIG. **12**, the projection **74**F is positioned at the second end **74**B of the expansion support bar **74** and extends outwardly from the exterior surface **74**D. Such inclusion of the projections **74**F may provide attachment means between the expansion foot **72** and the expansion support bar **74** when assembled with one another. Stated differently, expansion support bar **74** frictionally fits with the expansion foot **72** due to the projection **74**F pressing against an interior wall defining the central opening **72**J of the expansion foot **72**.
(61) Still referring to expansion support bar **74**, the expansion support bar **74** may also include an end identifier **74**G. As best seen in FIG. **12**, the end identifier **74** extends longitudinally from the first end **74**A towards a location that is defined between the first end **74**A and the second end **74**B. In one exemplary embodiment, the end identifier **74** may be shown as a series of arrows that extends longitudinally from the first end **74**A towards a location that is defined between the first end **74**A and the second end **74**B. In operation, the end identifier **74**G may indicate and/or signal to a user of the stand **1** and the expansion assembly **70** that the end of the expansions support bar **74** is near when moving the expansion assembly **70** away from the stand **1**. Such warning and/or indication prevents the user from accidently or incidentally removing the expansion support bar **74** from the support tube **30** when adjusting the expansion assembly **70**. It should be understood that additional measurement markings and/or indicia may be used to determine how close the first end **74**A is based on specific markings defining the end identifier **74**G.
(62) It should be understood that the expansion support tube may define any suitable length necessary for transporting and organizing surgical and/or medical equipment. In one exemplary embodiment, a length of an expansion support tube discussed herein may be approximately six inches long when measured between a first end of the support tube and a second end of the support tube. In another exemplary embodiment, another length of an expansion support tube discussed herein may be approximately twelve inches long when measured between a first end of the support tube and a second end of the support tube. In yet another exemplary embodiment, another length of an expansion support tube discussed herein may be between six inches up to about twelve inches when measured between a first end of the support tube and a second end of the support tube. In yet another exemplary embodiment, another length of an expansion support tube discussed herein may be at least twelve inches when measured between a first end of the support tube and a second end of the support tube.
(63) Having now described the components of the expansion assembly **70**, a method of assembling the expansion assembly **70** with the stand **1** is discussed in greater detail below.
(64) Initially, the user may retrieve the expansion foot **72** and the expansion support bar **74** to begin assembling the expansion assembly **70**. Upon assembly, the user introduces the second end **74**B of the expansion support bar **74** into the central opening **72**J of the expansion foot **72** to engage the expansion foot **72** and the expansion support bar **74** with one another. The projection **74**J of the expansion support bar **74** may also provide attachment means inside of the central opening **72**J of the expansion foot **72** to frictionally fit the expansion foot **72** and the expansion support bar **74** with one another.
(65) Once the expansion assembly **70** is built, the user may then assembly the expansion assembly **70** with the assembled stand **1** by inserting the first end **74**A of the expansion support bar **74** into the passageway **30**F of the support tube **30** at the first end **30**A or the second end **30**B. In one example, and as best seen in FIGS. **10-11**, the user inserts the first end **74**A of the expansion support bar **74** through the second end **30**B of the support tube **30** and into the passageway **30**F of the support tube **30**. Upon insertion of the expansion support bar **74**, the set of internal ridges **30**G

grips and engages with the exterior surface **74**D of the expansion support bar **74** to frictionally fit the expansion support bar **74** with the support tube **30** (see FIG. **11**). The user may continue to insert the expansion support bar **74** into the support tube **30** until a desired distance is met between the second foot **20** and the expansion foot **72**. The user may use and view through one of the apertures of the set of apertures **30**H defined in the support tube **30** to determine the position of the first end **74**A of the expansion support bar **74** (or at least a portion of the expansion support bar **74**) inside of the support tube **30**.

(66) Once the stand **1** and expansion assembly **70** are assembled with one another, user may introduce and equip one or more stringer bars **40** with the stand **1** and/or the expansion assembly **70** and one or more clips **50** with the stand **1** and/or the expansion assembly **70** to support and/or hold two sets of first surgical instruments **60**.

(67) In one embodiment, and as best seen in FIGS. **8-9**, the user may introduce the two sets of first surgical instruments **60** to the stand **1**. Once introduced, each surgical instrument of a first set of first surgical instruments **60**A illustrated herein may rest on and be supported by the support tube **30**, and each surgical instrument of a second set of first surgical instruments **60**B illustrated herein may rest on and be supported by the expansion support bar **74**. The user may then introduce one or more clips **50** to the stand **1** and to the expansion assembly **70** to organize and/or arrange the sets of first surgical instruments **60**A, **60**B in one or more desired groups. As best seen in FIGS. **8-9**, two clips **50** are shown being clamped to the support tube **30** at two locations between the first end **30**A and the second end **30**B to divide and/or separate the first set of first surgical instruments **60**A into three distinct groups while the second foot **20** separates the second set of first surgical instruments **60**B from the first set of first surgical instruments **60**A. It should be understood that the user may introduce and clamp the clips **50** with the support tube **30** or the expansion support bar **74** prior to introducing the sets of first surgical instrument **60**A, **60**B to the stand **1**.

(68) Once the clips **50** and the sets of first surgical instruments **60**A, **60**B are provided with stand **1**, user may then introduce and temporarily engage two stringer bars **40**′, **40**″ with the stand **1**, the expansion assembly **70**, and the sets of first surgical instruments **60**A, **60**B. As best seen in FIGS. **8-9**, the first end **40**A′ of the first stringer bar **40**′ may pass through the second accessory opening **20**K**2** of the pair of accessory openings **20**K of the second foot **20**. The first end **40**A′ of the first stringer bar **40**′ also passes through the second accessory opening **10**K**2** of the pair of accessory openings **10**K of the first foot **10** given the second accessory opening **10**K**2** of the pair of accessory openings **10**K of the first foot **10** and the second accessory opening **20**K**2** of the pair of accessory openings **20**K of the second foot **20** are coaxial with one another when stand **1** is assembled. The user continues to pass the first end **40**A′ of the first stringer bar **40**′ through the first foot **10** and the second foot **20** until the first portion **40**D′ is supported and engaged with the first foot **10** and the second foot **20**. As the first end **40**A′ passes through the first foot **10** and the second foot **20**, the second end **40**B′ of the stringer bar **40**′ simultaneously passes through second handles and/or grips **61**B of the first set of first surgical instruments **60**A so that the third portion **40**F′ hangs and supports the first set of first surgical instruments **60**A.

(69) Similarly, and as best seen in FIGS. **8-9**, the first end **40**A″ of the second stringer bar **40**″ may pass through a first accessory opening **72**K**1** of the pair of accessory openings **72**K of the expansion foot **72**. The first end **40**A″ of the second stringer bar **40**″ also passes through the first accessory opening **20**K**1** of the pair of accessory openings **20**K of the second foot **20** given the first accessory opening **20**K**1** of the pair of accessory openings **20**K of the second foot **20** and the first accessory opening **72**K**1** of the pair of accessory openings **72**K of the expansion foot **72** are coaxial with one another when stand **1** and expansion assembly **70** are assembled. The user continues to pass the first end **40**A″ of the second stringer bar **40**″ through the second foot **20** until the first portion **40**D″ of the second stringer bar **40**″ is supported and engaged with the second foot **20** and the expansion foot **72**. As the first end **40**A″ passes through the second foot **20** and the expansion foot **72**, the first end **40**A″ of the second stringer bar **40**″ simultaneously passes through first

handles and/or grips **61**A of the second set of first surgical instruments **60**B and first handles and/or grips **61**A of the first set of first surgical instruments **60**A so that the first portion **40**D″ supports some of the first set of first surgical instruments **60**A and the second set of first surgical instruments **60**B. As the first end **40**A″ passes through the second foot **20** and the expansion foot **72**, the second end **40**B″ of the second stringer bar **40**″ simultaneously passes through second handles and/or grips **61**B of the second set of first surgical instruments **60**B and second handles and/or grips **61**B of the first set of first surgical instruments **60**A so that the third portion **40**F″ hangs and supports some of the first set of first surgical instruments **60**A and the second set of first surgical instruments **60**B. Upon such engagement of the stand **1**, the stringer bars **40**′, **40**″, the clips **50**, the sets of first surgical instruments **60**, and the expansion assembly **70**, the user may then collectively transport the stand **1**, the stringer bar **40**, the clips **50**, the set of first surgical instruments **60**, and the expansion assembly **70** to desired locations discussed herein and other suitable locations found in hospitals or other medical buildings and offices.

(70) It should be understood that while two stringer bars **40** were used with the stand **1** and the expansion assembly **70**, any number of stringer bars may be equipped with just the stand **1**, a combination of the stand **1** and the expansion assembly **70**, a combination of the stand and at least two expansion assemblies **70**, and other combinations or configurations of the stand **1** discussed herein.

(71) FIGS. **13-16** illustrate a rack or second stand accessory, generally referred to as **80**, that operably engages with the stand **1** and/or expansion assembly **70** (if equipped). As best seen in FIG. **13**, the rack **80** is configured to engage with at least the first foot **10** and the second foot **20** and to support one or more types of surgical equipment or instruments at an elevated position that is above the support tube **30**. Such features and components of the rack **80** are discussed in greater detail below.

(72) As best seen in FIG. **16**, the rack **80** includes a top edge **80**A, a bottom edge **80**B that is vertically opposite to the top edge **80**A, and a vertical direction defined therebetween. Rack **80** also includes a first side edge **80**C that extends between the top edge **80**A and the bottom edge **80**B, and a second side edge **80**D that extends between the top edge **80**A and the bottom edge **80**B and is opposite to the first side edge **80**C. Rack **80** also defines an overall height **80**E that is measured between the top edge **80**A and the bottom edge **80**B, and an overall length **80**F that is measured between the first side edge **80**C and the second side edge **80**D. In the present disclosure, the overall length **80**F of the rack **80** is greater than the overall height **80**E and is equal to the overall length of the stand **1** measured between the outer sides **10**C, **20**C of the first foot **10** and the second foot **20**. In other exemplary embodiments, any rack discussed herein may have any suitable height or length as dictated by the implantation of said rack.

(73) Rack **80** also defines a pair of notches **80**G wherein a first notch **80**G′ is defined at the bottom edge **80**B near the first side edge **80**C, and a second notch **80**G″ is defined at the bottom edge **80**B near the second side edge **80**D. Each notch of the pair of notches **80**G is defined by a lower horizontal edge **80**G**1** that extends inwardly from the respective side edge **80**C, **80**D to a lower vertical edge **80**G**2**. Each notch of the pair of notches **80**G also defines a height **80**G**3** that is measured from the bottom edge **80**B to the respective lower horizontal edge **80**G**1**. Each notch of the pair of notches **80**G also defines a width **80**G**4** that is measured from the lower vertical edge **80**G**2** to the respective side edge **80**C, **80**D. Rack **80** also includes a pair of interior vertical edges **80**H where a first interior vertical edge **80**H**1** of the pair of interior vertical edges **80**H extends vertically upward from the bottom edge **80**B to an interior horizontal edge **80**J of the rack **80**. Similarly, a second interior vertical edge **80**H**2** of the pair of interior vertical edges **80**H extends vertically upward from the bottom edge **80**B to the interior horizontal edge **80**J of the rack **80**. In the present disclosure, the interior vertical edges **80**H and the interior horizontal edge **80**J collectively define a continuous U-shaped edge. Such U-shaped edge may become advantageous if a user would like to rest surgical instruments or devices on the support tube **30** without interference

or hindrance created by the rack **80**.
(74) Still referring to rack **80**, rack **80** also includes a pair of legs **80**K. As best seen in FIG. **16**, a first leg **80**K′ of the pair of legs **80**K is defined between the first side edge **80**C and the first interior vertical edge **80**H**1** and has a leg width **80**K**1**, and a second leg **80**K″ of the pair of legs **80**K is defined between the second side edge **80**D and the second interior vertical edge **80**H**2** and has the leg width **80**K**1**. In the present disclosure, the leg width **80**K**1** is greater than the width **80**G**4** of each notch of the pair of notches **80**G and the slit widths (e.g., **10**Q) defined by the first foot **10** and the second foot **20**.
(75) Still referring to rack **80**, rack **80** defines at least one group or set of recesses along the top edge **80**A of the rack **80**. As best seen in FIG. **16**, rack **80** defines a first group of recesses **80**L that extends downwardly into the rack **80** from the top edge **80**A towards the bottom edge **80**B. In one instance, a recess of the first group of recesses **80**L may be defined by a first notch height **80**L**1** and a first notch width **80**L**2** for receiving surgical instruments and/or devices that define a rectangular outer profile. In another instance, another recess of the first group of recesses **80**L may be defined by a first notch angle **80**L**3** for receiving surgical instruments and/or devices that define a circular outer profile and/or tapered outer profile. In the present disclosure, angle **80**L**3** is an acute angle for receiving surgical instruments and/or devices that define a circular outer profile and/or tapered outer profile. In other exemplary embodiments, angle **80**L**3** may be any suitable angle for receiving surgical instruments and/or devices that define a circular outer profile and/or tapered outer profile.
(76) Still referring to FIG. **16**, rack **80** may also define a second group of recesses **80**M that extends downwardly into the rack **80** from the top edge **80**A towards the bottom edge **80**B. In one instance, a recess of the second group of recesses **80**M may define a second notch height **80**M**1** that is less than the first notch height **80**L**1** of the first group of recesses **80**L. In this instance, notches of the second group of recesses **80**M may receive surgical instruments and/or devices that define smaller outer profiles as compared to the outer profiles of the surgical instruments and/or devices received by the first group of recesses **80**L.
(77) Still referring to FIG. **16**, rack **80** may also define a third group of recesses **80**N that extends downwardly into the rack **80** from the top edge **80**A towards the bottom edge **80**B. In one instance, a recess of the third group of recesses **80**N may be defined by a third notch height **80**N**1** and a second notch width **80**N**3** for receiving surgical instruments and/or devices that define a rectangular outer profile. In this instance, recesses of the third group of recesses **80**N may receive surgical instruments and/or devices that define smaller outer profiles as compared to the outer profiles of the surgical instruments and/or devices received by the first group of recesses **80**L and the second group of recesses **80**M.
(78) With such differences in size, shape, and configuration of the first group of recesses **80**L, the second group of recesses **80**M, and the third group of recesses **80**N, the rack **80** provides users and medical professionals to rest and/or hang various types of surgical instruments and devices from the rack **80** when such surgical instruments and devices are not being used during surgical and/or medical procedures. The rack **80** also places the surgical instruments and devices at an elevated state so that users and medical professionals may quickly grasp and remove a desired surgical instrument or device or quickly place and rest the used surgical instrument or device at the desired position on the rack **80**. The rack **80** also organizes various types of surgical instruments or devices based on the outer profiles or sizes of these surgical instruments or devices by placing and resting said surgical instruments into conforming recesses **80**L, **80**M, **80**N defined in rack **80**.
(79) While rack **80** defines a first group of recesses **80**L, a second group of recesses **80**M, and a third group of recesses **80**N, a rack discussed and illustrated herein may include any suitable number of notches for resting and/or hanging various types of surgical instruments or devices for a particular surgical or medical procedure. In one example, and as best seen in FIG. **19**, an alternative rack or second stand accessory **80′** may define a greater number recesses or slots so a user may rest and/or hang a greater number of surgical instruments or devices as compared to the rack **80**

discussed above. In this example, alternative rack **80′** defines an overall length **80**F′ that is greater than the overall length **80**F of the rack **80** discussed above. It should be noted that a user may desire to use alternative rack **80′** when stand **1** is equipped with the expansion assembly **70** since alternative rack **80′** is intended to engage with one of the first foot **10** and the second foot **20** of the stand **1** and the expansion foot **72** of the expansion assembly **70**.

(80) Still referring to rack **80**, rack **80** also defines a front surface **80**P and a rear surface **80**Q. As best seen in FIG. **15**, the front surface **80**P extends between the first side edge **80**C and the second side edge **80**D and is positioned ahead of the top edge **80**A, the bottom edge **80**B, the first side edge **80**C, and the second side edge **80**D. Still referring to FIG. **15**, the rear surface **80**Q also extends between the first side edge **80**C and the second side edge **80**D and is positioned behind the top edge **80**A, the bottom edge **80**B, the first side edge **80**C, and the second side edge **80**D. Rack **80** also defines an overall thickness **80**R that is continuous along the entire length of the rack **80** from the first side edge **80**C and the second side edge **80**D. In the present disclosure, the overall thickness **80**R of the rack **80** is less than the slit width **10**Q of the first foot **10** (as well as the second foot **20**) wherein the rack **80** frictionally fits inside of the first foot **10** and the second foot **20**.

(81) It should be understood that the rack **80** may define any suitable length necessary for transporting and organizing surgical and/or medical equipment. In one exemplary embodiment, a length of a rack discussed herein may be approximately six inches long when measured between a first side edge of the rack and a second side edge of the rack. In another exemplary embodiment, another length of a rack discussed herein may be approximately twelve inches long when measured between a first side edge of the rack and a second side edge of the rack. In yet another exemplary embodiment, another length of a rack discussed herein may be between six inches up to about twelve inches when measured between a first side edge of the rack and a second side edge of the rack. In yet another exemplary embodiment, another length of a rack discussed herein may be at least twelve inches when measured between a first side edge of the rack and a second side edge of the rack.

(82) Having now described the features of rack **80**, methods of assembling rack **80** with the stand **1** to rest and/or hang various types of surgical equipment from the rack **80** are discussed in greater detail below.

(83) Once the stand **1** is assembled, the user may then introduce and engage the rack **80** with the stand **1**. As best seen in FIG. **14**, the pair of notches **80**G of the rack **80** is received by the first slit **10**N of the first foot **10** and the second slit **20**N of the second foot **20**. The first slit **10**N of the first foot **10** and the second slit **20**N of the second foot **20** also receive portions of the rack **80** that are measured from the pair of notches **80**G to a medial location between the top edge **80**A and the pair of notches **80**G. Such portions of the rack **80** that engage with first foot **10** and the second foot **20** inside of the first slit **10**N and the second slit **20**N include front surface **80**P of the rack **80** and the rear surface **80**Q of the rack **80**. The lower vertical edges **80**G**2** that define the pair of notches **80**G may also engage with the inner sides **10**D, **20**D of the first foot **10** and the second foot **20**. Once the rack **80** is frictionally fit with the stand **1**, users may then rest and hang a set of second surgical instruments **90** inside of one or more of the first group of recesses **80**L, the second group of recesses **80**M, and the third group of recesses **80**N (see FIG. **13**).

(84) Rack **80** may also be operably engaged with the expansion assembly **70** if equipped with the stand **1** (see FIG. **18**). In this particular embodiment, the rack **80** would be received be the expansion slit **72**N of the expansion foot **72** to frictionally fit the rack **80** with the expansion foot **72** once the expansion assembly **70** is set to a desired distance relative to the first foot **10** or the second foot **20**.

(85) FIG. **17** illustrates another embodiment of the stand **1** that is equipped with the expansion assembly **70** and the rack **80** to hold both the set of first surgical instruments **60** and the set of second surgical instruments **90** discussed previously. In this embodiment, the set of first surgical

instruments **60** would rest on the expansion support bar **74** of the expansion assembly **70** while being separated by a single clip **50**. In this same embodiment, the set of second surgical instruments **90** would also rest on the rack **80** inside one or more of the first group of recesses **80**L, the second group of recesses **80**M, and the third group of recesses **80**M. While not illustrated, a user may engage at least one stringer bar **40** with the stand **1**, the expansion assembly **70**, and the set of first surgical instruments **60** if the user is collectively transporting the stand **1**, the expansion assembly **70**, and the set of first surgical instruments **60** from an in-queue station to a surgical station or sanitization station.

(86) It should be understood that combination of the stand **1** and the expansion assembly **70** may define any suitable length necessary for transporting and organizing surgical and/or medical equipment. In one exemplary embodiment, a length of combination of a stand and an expansion assembly discussed herein may be approximately six inches long when measured between a first side edge of the plate and a second side edge of the plate. In another exemplary embodiment, another length of a plate of a clamping rack discussed herein may be approximately twelve inches long when measured between a first side edge of the plate and a second side edge of the plate. In yet another exemplary embodiment, another length of a plate of a clamping rack discussed herein may be between six inches up to about twelve inches when measured between a first side edge of the plate and a second side edge of the plate. In yet another exemplary embodiment, another length of a plate of a clamping rack discussed herein may be at least twelve inches when measured between a first side edge of the plate and a second side edge of the plate.

(87) FIG. **20** illustrates another embodiment of the stand **1** that is equipped with a first expansion assembly **70**A, a second expansion assembly **70**B, and a rack **80**. In this particular embodiment, an expansion support bar **74** of the first expansion assembly **70**A may be inserted into the support tube **30** at the first end **30**A to slidably engage the first expansion assembly **70**A with the stand **1**. In this same embodiment, an expansion support bar **74** of the second expansion assembly **70**B may be inserted into the support tube **30** at the second end **30**B to slidably engage the second expansion assembly **70**B with the stand **1**. Such configuration may be desirable by a user when large amounts of surgical instruments (such as the set of first surgical instruments **60**) must be transported for surgical needs or sterilization needs, various types of surgical instrument or devices (such as the set of first surgical instruments **60** and set of second surgical instruments **90**) are needed during a single surgical procedure, and other various considerations of the like.

(88) FIGS. **21-25** illustrates another embodiment of the stand **1** that is equipped with at least one clamping rack **100**. As discussed in greater detail below, the at least one clamping rack **100** is selectively operably engageable with the support tube **30** between the first foot **10** and the second foot **20**. Such features and components of the clamping rack **100** are discussed in greater detail below.

(89) The clamping rack **100** includes a plate **102**. In the present disclosure, certain features of the plate **102** of the clamping plate **102** are identical to certain features of the rack **80** mentioned above. As such, a top edge **102**A, a bottom edge **102**B, a first side edge **102**C, a second side edge **102**D, a pair of interior vertical edges **102**H, a pair of interior horizontal edges **102**J, a pair of legs **102**K, at least one group of recesses **102**L, a front surface **102**P, and a rear surface **102**Q of the plate **102** of the clamping rack **100** are identical to the top edge **80**A, the bottom edge **80**B, the first side edge **80**C, the second side edge **80**D, the pair of interior vertical edges **80**H, the pair of interior horizontal edges **80**J, the pair of legs **80**K, a group of recesses **80**L, a front surface **80**P, and a rear surface **80**Q of the rack **80**.

(90) However, the plate **102** of the clamping rack **100** includes different features and/or configurations when compared to the rack **80**. As best seen in FIG. **24**, the plate **102** includes a pair of cavities **102**R that extends into each leg of the pair of legs **102**K. In the present disclosure, a first cavity of the pair of cavities **102**R extends into each leg of the pair of legs **102**K from the first side edge **102**C to a respective interior vertical edge of the pair of interior vertical edges **102**H. In the

present disclosure, a second cavity of the pair of cavities **102**R also extends into each leg of the pair of legs **102**K from a respective interior vertical edge of the pair of interior vertical edges **102**H to the first side edge **102**C in which the second cavity of the pair of cavities **102**R opposes the first cavity of the pair of cavities **102**R. Such use and purpose of the pair of cavities **102**R defined in each leg of the pair of legs **102**K is discussed in greater detail below.
(91) The plate **102** of the clamping rack **100** also includes a pair of attachment arms **102**S that is provided on each leg of the pair of legs **102**K. As best seen in FIG. **24**, each attachment arm of the pair of attachment arms **102**S extends from the respective leg of the pair of legs **102**K and is positioned forwardly of the front surface **102**P of the plate **102**. In one exemplary embodiment, each attachment arm of the pair of attachment arms **102**S is bent forwardly from the respective leg of the pair of legs **102**K which defines the pair of cavities **102**R in each leg of the pair of legs **102**K. Each attachment arm of the pair of attachment arms **102**S also defines an opening **102**T that extends transversely through each attachment arm of the pair of attachment arms **102**S. Such use and purpose of the pair of attachment arms **102**S provided on each leg of the pair of legs **102**K is discussed in greater detail below.
(92) The plate **102** of the clamping rack **100** also includes a first clamping surface **102**U. As best seen in FIGS. **24** and **25**, the first clamping surface **102**U is a section of the front surface **102**P defined along each leg of the pair of legs **102**K. Particularly, the first clamping surface **102**U is defined along each leg of the pair of legs **102**K between the bottom edge **102**B and the pair of attachment arms **102**S. In one instance, the first clamping surface **102**U engages with the exterior surface **30**D of the support tube **30** when the clamping rack **100** engages with the support tube **30**. In another instance, the first clamping surface **102**U may engage with the exterior surface **74**D of the expansion support bar **74** when the clamping rack **100** engages with the expansion support bar **74**.
(93) In the present disclosure, the first clamping surface **102**U defined along each leg of the pair of legs **102**K is planar and/or flat which matches the front surface **102**P of the plate **102**. In one exemplary embodiment, the first clamping surface **102**U defined along each leg of the pair of legs **102**K may be curvilinear and/or non-planar. In this exemplary embodiment, the first clamping surface **102**U defined along each leg of the pair of legs **102**K may match the exterior surface **30**D of the support tube **30** and/or the exterior surface **74**D of the expansion support bar **74**.
(94) It should be understood that the plate **102** of clamping rack **100** may define any suitable length necessary for transporting and organizing surgical and/or medical equipment. In one exemplary embodiment, a length of a plate of a clamping rack discussed herein may be approximately six inches long when measured between a first side edge of the plate and a second side edge of the plate. In another exemplary embodiment, another length of a plate of a clamping rack discussed herein may be approximately twelve inches long when measured between a first side edge of the plate and a second side edge of the plate. In yet another exemplary embodiment, another length of a plate of a clamping rack discussed herein may be between six inches up to about twelve inches when measured between a first side edge of the plate and a second side edge of the plate. In yet another exemplary embodiment, another length of a plate of a clamping rack discussed herein may be at least twelve inches when measured between a first side edge of the plate and a second side edge of the plate.
(95) The clamping rack **100** may also include a set of clips **104** that pivotably engages with the plate **102**. As best seen in FIGS. **22-24**, each clip of the set of clips **104** includes a top end **104**A, a bottom end **104**B vertically opposite to the top end **104**A, a front surface **104**C that extends between top end **104**A and the bottom end **104**B and faces away from the plate **102**, and a rear surface **104**D that extends between the top end **104**A and the bottom end **104**B and faces the plate **102**.
(96) Each clip of the set of clips **104** also defines a pair of cavities **104**E. As best seen in FIG. **23**, the pair of cavities **104**E extends forwardly from the rear surface **104**D towards the front surface

104C to receive a respective pair of attachment arms **102**S of the plate **102** (see FIG. **25**). Each clip of the set of clips **104** also includes a pair of attachment extensions **104**F that extends rearward from rear surface **104**D. Upon assembly of the clamping rack **100**, the pair of attachment extensions **104**F is positioned internally of a respective pair of attachment arms **102**S of the plate **102** to pivotably engage each clip of the set of clips **104** with the plate **102**; such pivot engagement is discussed in greater detail below. Each attachment extension of the pair of attachment extensions **104**F also defines an opening **104**G that extends transversely through each attachment extension of the pair of attachment extension **104**G. Such use and purpose of the pair of attachment extensions **104**E provided on each clip **104** is discussed in greater detail below.

(97) Each clip of set of clips **104** also defines a grip **104**H. As best seen in FIGS. **21-25**, the grip **104**H is defined along the front surface **104**C of each clip of the set of clips **104** at the top end **104**A. In the present disclosure, the grip **104**H is a plurality of ridges and/or bumps that extends upward from the front surface **104**C. In other exemplary embodiments, grip **104**H may be defined by any suitable members and/or features on each clip of the set of clips **104**.

(98) Each clip of the set of clips **104** also includes a second clamping surface **104**J. As best seen in FIGS. **23-25**, the second clamping surface **104**J is a section of the rear surface **104**D defined along each clip of the set of clips **104**. Particularly, the second clamping surface **104**J of each clip **104** is defined between the bottom end **104**B and the pair of attachment extensions **104**F. In one instance, the second clamping surface **104**J engages with the exterior surface **30**D of the support tube **30** when the clamping rack **100** engages with the support tube **30**. In another instance, the second clamping surface **104**J may engage with the exterior surface **74**D of the expansion support bar **74** when the clamping rack **100** engages with the expansion support bar **74**. In the present disclosure, the second clamping surface **104**J of each clip of the set of clips **104** also defines a curvilinear shape when viewed from a sectional view such that the second clamping surface **104**J matches with the exterior surface **30**D of the support tube **30** and the exterior surface **74**D of the expansion support bar **74** (see FIG. **25**).

(99) In other exemplary embodiments, the second clamping surface **104**J of each clip of the set of clips **104** may include additional features and/or components to prevent axial rotational of the clamping rack **100** when engaged with the support tube **30** or the expansion support rod **74**. In one exemplary embodiment, a set of teeth may be defined along the second clamping surface **104**J of each clip of the set of clips **104**. Such inclusion of the set of teeth provides additional grip between each clip of the set of clips **104** and one of the support tube **30** and the expansion support bar **74**.

(100) While not illustrated herein, each clip of the set of clips **104** may define a through-hole that extends between the front surface **104**C and the second clamping surface **104**J. Such inclusion of a through-hole in each clip of the set of clips **104** may allow fluid to pass through the respective clip **104** when the clamping rack **100** is being cleaned and remains engaged with the stand **1** or the expansion assembly **70**. It should also be understood that a through-hole may also be include in other clips and similar clamping devices mentioned herein.

(101) Clamp assembly **100** also includes pivot pin **106** that pivotably engages the plate **102** and the clips **104** with one another. As best seen in FIG. **25**, the pivot pin **106** operably engages with the pairs of attachment arms **102**S inside of the openings **102**T and with the pairs of attachment extensions **104**F inside of the openings **104**G. In the present disclosure, the plate **102** and/or clip **104** rotates about the pivot pin **106** so that the clamping rack **100** may be selectively engaged with the support tube **30** or at least one extension support bar **74**.

(102) Still referring to FIG. **25**, clamping assembly **100** also includes a biaser **108** that operably engages between the plate **102** and a clip of the set of clips **104**. In the present disclosure, the biaser **108** is configured to apply an inward force on the plate **102** and the respective clip **104** such that the first clamping surface **102**U of the plate **102** and the second clamping surface **104**J of the respective clip **104** are biased towards one another. Such configuration of each biaser **108** enables the plate **102** and the set of clips **104** to apply a constant inward pressure towards one another so

that the clamping rack **100** remains engaged with the support tube **30** or the extension support bar **74**.

(103) It should be understood that any suitable components and/or mechanism may be used to pivotably engage the plate **102** and the clips **104** with one another. In one exemplary embodiment, a fastener and a rivet nut may be provided to pivotably engage the plate **102** and the clips **104** with one another. In this exemplary embodiment, the plate **102** and/or clip **104** rotates about the rivet nut so that the clamping rack **100** may be selectively engaged with the support tube **30** or at least one extension support bar **74**. It should be understood that any suitable biaser or biasing means may be used herein to apply an inward force on the plate **102** and the respective clip **104** such that the first clamping surface **102**U of the plate **102** and the second clamping surface **104**J of the respective clip **104** are biased towards one another. In one exemplary embodiment, a torsion spring operably engages with the plate **102** and a respective clip of the set of clips **104** that applies an inward force on the plate **102** and the respective clip **104** such that the first clamping surface **102**U of the plate **102** and the second clamping surface **104**J of the respective clip **104** are biased towards one another.

(104) FIG. **26** illustrates a kit **200**. In the illustrated embodiment, kit **200** may include various components and/or accessories that are mentioned herein. In the present embodiment, the kit **200** may include components of a stand **202**, particularly a first foot **202**A (e.g., first foot **10** of stand **1**), a second foot **202**B (e.g., second foot **20** of stand **1**), and support tube **202**C (e.g., support tube **30** of stand **1**). The kit **200** may also include at least two or more stringer bars **204** (e.g., stringer bars **40**) and a plurality of clips **206** (e.g., clips **50**). The kit **200** may also include a rack **208** (e.g., rack **80**) and at least one expansion assembly **110**, particularly an expansion foot **210**A (e.g., expansion foot **72** of expansion assembly **70**) and an expansion support rod **210**B (e.g., expansion support bar **74** of expansion assembly **70**). The kit **200** may also include a clamping rack **212** (e.g., clamping rack **100**) that is discussed and illustrated herein. It should be understood that kit **200** may include alternative components that are discussed herein as well as additional components that are currently included in kit **200**.

(105) FIG. **27** illustrates a method **300** of supporting at least one set of surgical instruments with a surgical instrument stand apparatus. An initial step **302** of method **300** includes engaging a first foot of the surgical instrument stand apparatus with a first end of a support tube of the surgical instrument stand apparatus. Another step **304** of method **300** includes engaging a second foot of the surgical instrument stand apparatus with a second end of the support tube, wherein the second foot is spaced apart from the first foot at a fixed distance defined by the support tube. Another step **306** of method **300** includes selecting between a first stand accessory and a second stand accessory. Another step **308** of method **300** includes engaging the selected first stand accessory or the second stand accessory with the first foot and the second foot. Another step **210** of method **300** includes supporting at least one set of surgical instruments with the surgical instrument stand apparatus by the support tube and the selected first stand accessory or the selected second stand accessory.

(106) In other exemplary embodiments, method **300** may include additional and/or optional steps. In one exemplary embodiment, method **300** may include that when the first stand accessory is selected, the method further comprises: inserting a first end of the first stand accessory through the first foot, the second foot, and first handles of the at least one set of surgical instruments; engaging the first stand accessory with the first foot, the second foot, and the first handles of the at least one set of surgical instruments; inserting a second end of the first stand accessory through second handles of the at least one set of surgical instruments; and engaging the first stand accessory with the second handles of the at least one set of surgical instruments. In another exemplary embodiment, method **300** may include steps of clamping at least one clip with the support tube; and separating the at least one set of surgical instruments into at least two groups. In another exemplary embodiment, method **300** may include that when the second stand accessory is selected in the step of selecting between the first stand accessory and the second stand accessory, the method further

comprises: engaging a first end of the second stand accessory with the first foot inside a first slit defined in the first foot; and engaging a second end of the second stand accessory with the second foot inside a second slit defined in the second foot; wherein the second stand accessory is positioned above and spaced apart from the support tube. In another exemplary embodiment, method **300** may include steps of engaging at least one expansion assembly with the support tube at one or both of the first end of the support tube and the second end of the support tube. In another exemplary embodiment, method **300** may include that when the second stand accessory is selected in the step of selecting between the first stand accessory and the second stand accessory, the method further comprises: engaging a first end of the second stand accessory with one of the first foot inside a first slit defined in the first foot and the second foot inside a second slit defined in the second foot; and engaging a second end of the second stand accessory with an expansion foot of the expansion assembly inside an expansion slit defined in the expansion foot; wherein the second stand accessory is positioned above and spaced apart from the support tube and an expansion support bar of the expansion assembly.

(107) It should be understood that the term “stand accessory” or similar terms used herein may be any accessory that is capable and/or equipped to support and/or organize one or more sets of surgical instruments on the stand **1** or at least one expansion assembly **70**. In one instance, a stand accessory may be stringer bar **40** that is capable of supporting and/or organizing one or more sets of surgical instruments on the stand **1** or at least one expansion assembly **70**. In another instance, a stand accessory may also be racks **80**, **80′** that is capable of supporting and/or organizing one or more sets of surgical instruments on the stand **1** or at least one expansion assembly **70**. In yet another instance, a stand accessory may clamping rack **100** that is capable of supporting and/or organizing one or more sets of surgical instruments on the stand **1** or at least one expansion assembly **70**.

(108) Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

(109) While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

(110) The articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims (if at all), should be understood to mean “either or

both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

(111) As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

(112) While components of the present disclosure are described herein in relation to each other, it is possible for one of the components disclosed herein to include inventive subject matter, if claimed alone or used alone. In keeping with the above example, if the disclosed embodiments teach the features of A and B, then there may be inventive subject matter in the combination of A and B, A alone, or B alone, unless otherwise stated herein.

(113) As used herein in the specification and in the claims, the term “effecting” or a phrase or claim element beginning with the term “effecting” should be understood to mean to cause something to happen or to bring something about. For example, effecting an event to occur may be caused by actions of a first party even though a second party actually performed the event or had the event occur to the second party. Stated otherwise, effecting refers to one party giving another party the tools, objects, or resources to cause an event to occur. Thus, in this example a claim element of “effecting an event to occur” would mean that a first party is giving a second party the tools or resources needed for the second party to perform the event, however the affirmative single action is the responsibility of the first party to provide the tools or resources to cause said event to occur.

(114) When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature

or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

(115) Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”, “behind”, “in front of”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

(116) Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first feature/element without departing from the teachings of the present invention.

(117) An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments.

(118) If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

(119) As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is +/−0.1% of the stated value (or range of values), +/−1% of the stated value (or range of values), +/−2% of the stated value (or range of values), +/−5% of the stated value (or range of values), +/−10% of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein.

(120) Additionally, the method of performing the present disclosure may occur in a sequence different than those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.
(121) In the claims, as well as in the specification above, all transitional phrases such as "comprising," "including," "carrying," "having," "containing," "involving," "holding," "composed of," and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases "consisting of" and "consisting essentially of" shall be closed or semi-closed transitional phrases, respectively.
(122) To the extent that the present disclosure has utilized the term "invention" in various titles or sections of this specification, this term was included as required by the formatting requirements of word document submissions pursuant the guidelines/requirements of the United States Patent and Trademark Office and shall not, in any manner, be considered a disavowal of any subject matter.
(123) In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.
(124) Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

Claims

1. A surgical instrument stand apparatus, comprising: a first foot; a second foot; a support tube extending between the first foot and the second foot and being a unitary, monolithic member, the support tube having a first end operably engaging with the first foot, a second end longitudinally opposite to the first end and operably engaging with the second foot, and a fixed length measured between the first end and the second end, wherein the first foot and the second foot are held at a fixed distance from one another by the support tube; and at least one stand accessory selectively operably engageable with the first foot and the second foot for supporting at least one set of surgical instruments.
2. The surgical instrument stand apparatus of claim 1, wherein when the at least one set of surgical instruments is supported by the at least one stand accessory, the at least one stand accessory is engaged with the first foot and the second foot at first positions.
3. The apparatus of claim 1, wherein each of the first foot and the second foot comprises: an outer side; an inner side facing in an opposite direction relative to the outer side; a central opening extending between the outer side and the inner side and configured to receive the support tube; and at least two accessory openings extending between the outer side and the inner side and being offset from the central opening; and wherein one of the at least two accessory openings is configured to receive the at least one stand accessory; and wherein the at least two accessory openings of the first foot and the second foot define diameters that are equal to one another.
4. The apparatus of claim 1, wherein the support tube further comprises: a wall extending between the first end and the second end and defining a passageway therethrough with an inner diameter.
5. The apparatus of claim 4, wherein the support tube further comprises: at least one opening defined in the wall between the first end of the support tube and the second end of the support tube; wherein the at least one opening provides fluid communication between the passageway of the support tube and an external environment that surrounds the support tube.
6. The apparatus of claim 4, further comprising: an expansion assembly selectively operably engagable with the support tube at one of the first end of the support tube and the second end of the support tube.
7. The apparatus of claim 6, wherein the expansion assembly comprises: an expansion foot; and an

expansion support bar extending between the expansion foot and one of the first foot and the second foot.
8. The apparatus of claim 7, wherein the expansion support bar comprises: a first end that operably engages with the expansion foot; a second end that is longitudinally opposite to the first end and is configured to selectively operably engage with the support tube at one of the first end of the support tube and the second end of the support tube; and a wall extending between the first end and the second end and defining an outer diameter that is less than the inner diameter of the support tube; wherein the expansion support bar is slidably moveably inside of the passageway of the support tube.
9. The apparatus of claim 1, further comprising: at least one clip that is selectively operably engageable with the support tube; wherein the at least one clip is selectively positionable along the support tube between the first foot and the second foot, wherein the at least one clip is configured to separate one surgical instrument of the at least one set of surgical instruments from another surgical instrument of the at least one set of surgical instruments.